

Phase 02 — Trial design · Pictorial walkthrough (D1: Sample-size + SAP drafter)

Phase context. Once the evidence base supports a new trial, the team specifies the question (PICOT), powers it (sample size), and writes the analysis plan. Deliverable: an ICH-E9-shaped **Statistical Analysis Plan** (PDF + DOCX), often submitted to peer review before any subject enrolls.

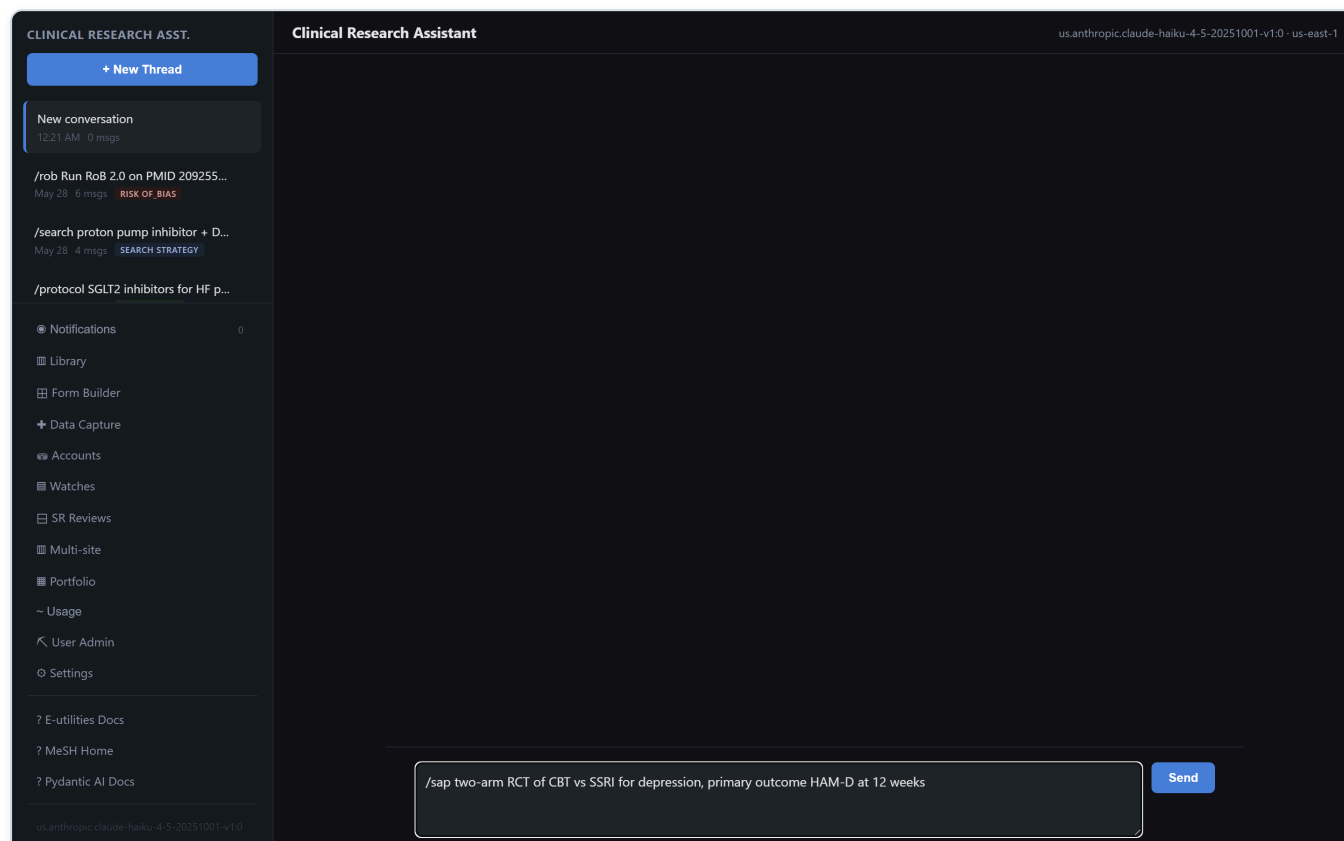
This guide was captured against a **live instance** (<http://localhost:8000> , AWS Bedrock `claude-haiku-4-5`) on 2026-06-19. Every screenshot below is a real render of the running app.

The workflow has four stages: **PICOT intake** → **sample-size derivation** → **ICH-E9 analysis plan** → **assembled SAP**. The `sample_size` calculator runs host-side on `scipy.stats` (no sandbox round-trip); every other card is produced by the `sap_drafter` specialist on Bedrock.

Step 1 — New conversation, trigger the SAP workflow

Click + **New Thread**, then paste the trigger into the composer and **Send**:

```
/sap two-arm RCT of CBT vs SSRI for depression, primary outcome HAM-D at 12 weeks
```



Step 2 — Review the PICOT intake card (STEP 1)

The model parses the trigger into a structured **PICOT** card — Population, Intervention, Comparator, Primary outcome, Timeframe, plus design / hypothesis / outcome-type. Review and edit inline if needed.

PICOT

STEP 1 — review the trial-design intake; confirm to proceed

POPULATION Adults with moderate-to-severe depression (HAM-D ≥ 17)	INTERVENTION Cognitive Behavioral Therapy (CBT)	COMPARATOR SSRI (selective serotonin reuptake inhibitor)	PRIMARY OUTCOME Hamilton Depression Rating Scale (HAM-D) score change from baseline to 12 weeks
TIMEFRAME Primary outcome assessment at 12 weeks; total follow-up 12 weeks	DESIGN parallel rct	HYPOTHESIS superiority	OUTCOME TYPE continuous

PICOT confirmed → STEP 2

Step 3 — Confirm PICOT and supply the statistical assumptions (STEP 2)

The drafter will **not invent an effect size** — a deliberate anti-hallucination guardrail. So instead of just clicking *PICOT confirmed*, answer with the confirmation **and** the assumptions in one message:

```
PICOT confirmed. For STEP 2 use a two-sample t-test: control HAM-D mean 18,
intervention mean 14 (mean difference 4), common SD 7, alpha 0.05, power 0.90,
allocation 1:1, 15% dropout.
```

The `sample_size` tool computes **N = 77 per arm, 154 total** (Cohen two-sample t-test), echoing back every assumption so the derivation is auditable. Click **Sample size confirmed → STEP 3**.

SAMPLE_SIZE

STEP 2 — required N from the sample_size tool

N PER ARM (CONTROL)

77

N PER ARM (INTERVENTION)

77

TOTAL ENROLMENT

154

FORMULA

Two-sample t-test (Cohen, common SD)

Cohen J. Statistical Power Analysis for the Behavioral Sciences, 2nd ed. Erlbaum, 1988. §2.4.

ASSUMPTIONS

```
{
  "control_event_rate": null,
  "relative_risk_reduction": null,
  "absolute_risk_difference": null,
  "control_mean": 18,
  "intervention_mean": 14,
  "standard_deviation": 7,
  "hazard_ratio": null,
  "control_survival_at_horizon": null,
  "mean_difference": null,
  "sd_of_difference": null,
  "alpha": 0.05,
  "power": 0.9,
  "allocation_ratio": 1,
  "dropout_rate": 0.15,
  "one_sided": false,
  "non_inferiority_margin": null
}
```

Sample size confirmed → STEP 3

Step 4 — Review the ICH-E9 analysis plan (STEP 3)

The drafter produces the methodological core: analysis populations (ITT / Safety), primary test (ANCOVA on HAM-D with baseline covariate), multiplicity strategy, missing-data handling, plus sensitivity / subgroup analyses and safety monitoring. Click **Analysis plan confirmed** → **STEP 4**.

ANALYSIS_PLAN

STEP 3 — ICH E9 methodological core

POPULATIONS

ITT / Safety

PRIMARY TEST

Two-sample t-test
(ANCOVA with baseline
HAM-D covariate)

MULTIPLICITY

none

MISSING DATA

multiple imputation
mar

PRIMARY ANALYSIS DESCRIPTION

Comparison of HAM-D score change from baseline to 12 weeks between CBT and SSRI arms using a two-sample t-test, with baseline HAM-D score as a covariate. Analysis conducted on the intention-to-treat population.

SENSITIVITY ANALYSES

- Per-protocol analysis excluding significant protocol violators
- On-treatment analysis censoring follow-up after treatment discontinuation
- Tipping-point sensitivity analysis for missing data under MNAR assumptions

SUBGROUP ANALYSES

- Age group (≤ 40 vs > 40 years)
- Baseline HAM-D severity (17–23 vs ≥ 24)
- Comorbid anxiety disorder (yes vs no)

SAFETY MONITORING

Safety population (all randomized participants who receive at least one dose) will be monitored for adverse events (AE) and serious adverse events (SAE). A Data Monitoring Committee will review SAE reports monthly. Stopping rules for safety: trial halted if ≥ 3 unexpected serious adverse reactions attributable to study treatment are reported within any 4-week period.

Analysis plan confirmed → STEP 4

Step 5 — Assemble the SAP (STEP 4), then finalize

The assembled **SAP document** card appears in draft state with a grounded background paragraph, references, and a *Show full Markdown* expander. Iterate with free-form messages, or click **Finalize SAP** to lock it.

SAP_DOCUMENT

FINAL — ready to download

BACKGROUND

Major depressive disorder (MDD) remains a leading cause of disability worldwide, with substantial heterogeneity in treatment response. Both cognitive behavioral therapy (CBT) and selective serotonin reuptake inhibitors (SSRIs) are established first-line treatments; however, head-to-head comparison of their efficacy on symptom severity change is limited. This parallel-group, two-arm randomized controlled trial aims to test the hypothesis that CBT is superior to SSRI monotherapy in reducing depressive symptoms (measured by the Hamilton Depression Rating Scale) at 12 weeks post-randomization in adults with moderate-to-severe major depressive disorder. The Hamilton Depression Rating Scale (HAM-D) is a well-validated clinician-administered outcome measure with established reliability and sensitivity to change in depression severity.

► Show full Markdown

REFERENCES

1. Statistical Power Analysis for the Behavioral Sciences — Cohen J

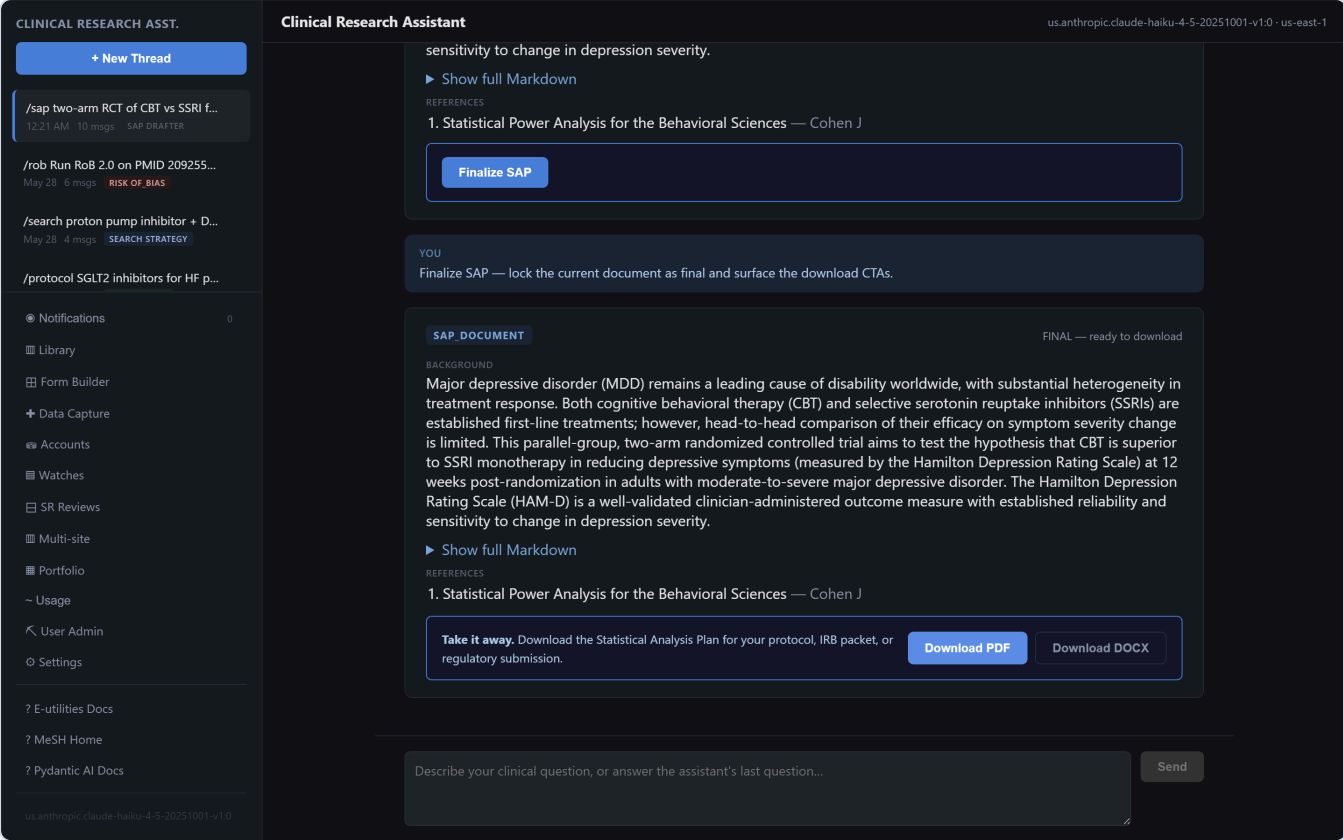
Take it away. Download the Statistical Analysis Plan for your protocol, IRB packet, or regulatory submission.

Download PDF

Download DOCX

Step 6 — Download the report

The finalized card surfaces **Download PDF / Download DOCX**. The PDF (`phase02-sap/sap-report.pdf`) bundles: Title · PICOT · Sample size · Analysis plan · Methods · Assumptions.



Sanity checks (acceptance criteria)

Check	Expected	Observed
Sample-size shape	<code>N_per_arm</code> , <code>total_n</code> , dropout-inflated N, formula, assumptions echoed	✔ 77 / 77 / 154, Cohen two-sample t-test
Effect size never fabricated	Drafter asks for assumptions rather than inventing them	✔ supplied by operator
Estimands / ICH-E9 framing	Populations + primary test + multiplicity + missing-data on the analysis card	✔ ITT/Safety, ANCOVA, multiple imputation
Downloadable artefact	PDF + DOCX from the final card	✔ valid 2-page %PDF-1.4

Variant to narrate: the four superiority formulas (two-proportions / two-means / Schoenfeld time-to-event / paired) ship today; non-inferiority + equivalence are roadmap items.

Reproduce

```
uv run python scripts/demo_capture_d1_sap.py # re-captures every screenshot in phase02-sap/
```